AMPERESAND · TECHNICAL INTERVIEW

Sai Kiran Reddy, Konda

Introduction → Education → Career → Technical Presentation → Q&A

Role: Engineer **Focus:** Distributed Systems & Cloud Architecture

01 · EDUCATION

Hardware roots, software ambitions

From analog circuits to distributed cloud systems.

Anna University

B.E. — Electronics & Communication Engineering

Microprocessors • Embedded Systems • Digital Electronics • Instrumentation

Northeastern University

M.S. — Computer Software Engineering

Algorithms • Database Design • Scalable Cloud Infrastructure • DevOps

The bridge: Electronics gave me physical systems thinking. Software gave me scale. The intersection — bridging hardware constraints with cloud infrastructure — is where I do my best work.

From mainframes to microservices

COMPANY	ROLE	WHAT I DID
Cognizant / Lincoln Financial	Project Lead	Migrated Federal Insurance from IBM Mainframe & S2K to ETL pipelines. 4 teams, 3 time zones, COVID. Delivered in 6 months.
Digital HIE Inc	Technical Lead	Led DirectTrust HISP accreditation from zero. HIPAA compliance, platform architecture, encryption, interoperability. 4 months.
Capital One	Systems Architect	Redesigned Transaction & Account Stores for 130M+ accounts. Monolith to serverless on AWS. Terraform, Go, DynamoDB.

Pattern: Take ownership of hard problems early, deliver under pressure, build systems others adopt.

03 · TECHNICAL INTERESTS

Where hardware meets scalable software

- **Architecture & System Design** — distributed systems at enterprise scale
- **Multi-Cloud Tenancies** — bare metal + cloud provider infrastructure
- **Bridging Physical Systems** with scalable software layers
- **Cryptography & Hashing** — security at the protocol level

> **Why Amperesand:** Solid-state transformers bridging physical power systems with intelligent software. That hardware-software intersection is exactly where I've built my career.

IaaS Architecture

Data Center

Virtualization

Servers

Storage

Networking

Security

TECHNICAL PRESENTATION

Capital One: Monolith to Microservices

Redesigning Transaction & Account Stores for 130M+ accounts across Card, Bank, Auto, and Global Finance.

Duration: 8 months **Scale:** 130M+ accounts **Stack:** Go, DynamoDB, Lambda, Terraform

One monolith. Four lines of business. It couldn't scale.

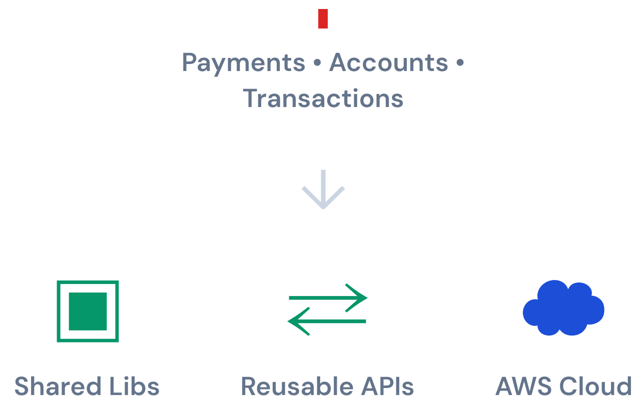
Capital One's Transaction and Account systems ran as a **single monolithic application**. Card, Bank, Auto, and Global Finance shared the same codebase, database, and deployment pipeline.

Changes to one LOB risked breaking another. Deployments were slow. Performance bottlenecks cascaded everywhere.

> Split the monolith into individual services with shared libraries and reusable APIs. Deploy on AWS. Without disrupting 130M+ accounts.

Before vs After

BEFORE	AFTER
Monolith Single app, all LOBs	Card Service Independent
Shared DB One schema	Bank Service Own model
Coupled Change one, risk all	Auto Service Isolated
Bottleneck Cascading	Global Finance Scales alone



Product requirements drive every decision

STEP	WHAT WE DID	DECISION
01 — Requirements	Mapped every account type and transaction flow. Stakeholders: Product Owners, Vendors, Architects.	Account ID & Transaction ID schema
02 — Schema	Designed with room for future changes without breaking migration.	Flexible document model
03 — Database	SQL vs NoSQL against current and future scale. Millions of concurrent ops.	NoSQL — DynamoDB
04 — Language	Selected for 15+ year horizon, performance, concurrency.	Go
05 — Cloud	Containers (k8s) vs serverless against cost and ops complexity.	Serverless — Lambda

Key insight: The default approach is containers + Kubernetes. We challenged that — serverless was more cost-effective at this scale and eliminated operational overhead.

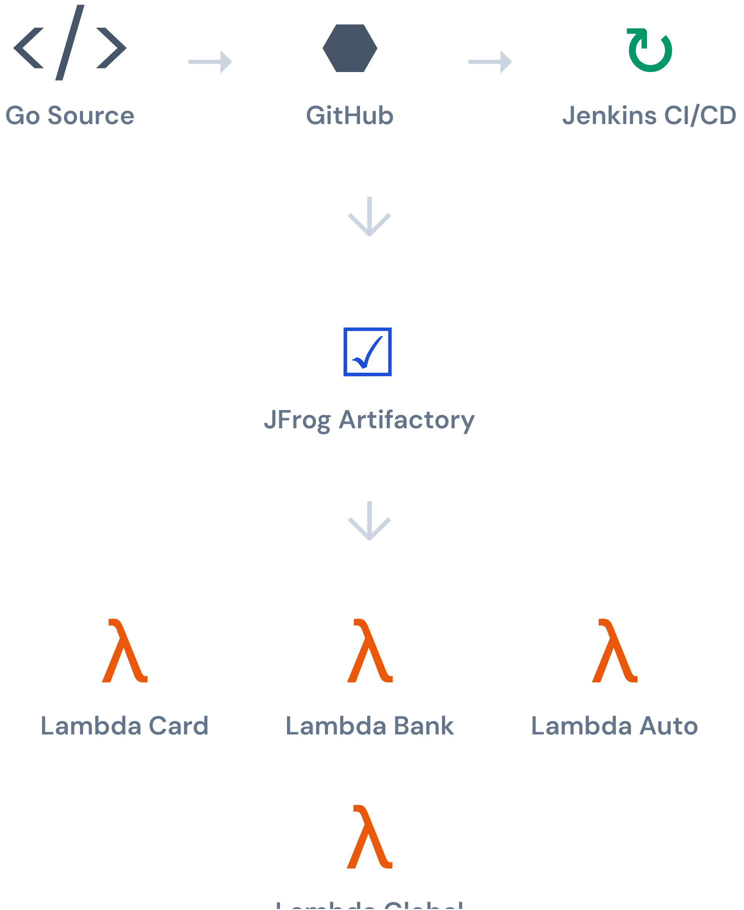
Request flow: client to DynamoDB and back

Users → Security → Cloud Doctor → Load Balancer → SQS → Lambda → DynamoDB → Monitoring → Cost Analysis

Ship fast. Test everything.

CI/CD Pipeline

Source code in **Go**, stored in GitHub. Jenkins builds binaries, pushes to **JFrog**, deploys to all lines of business.

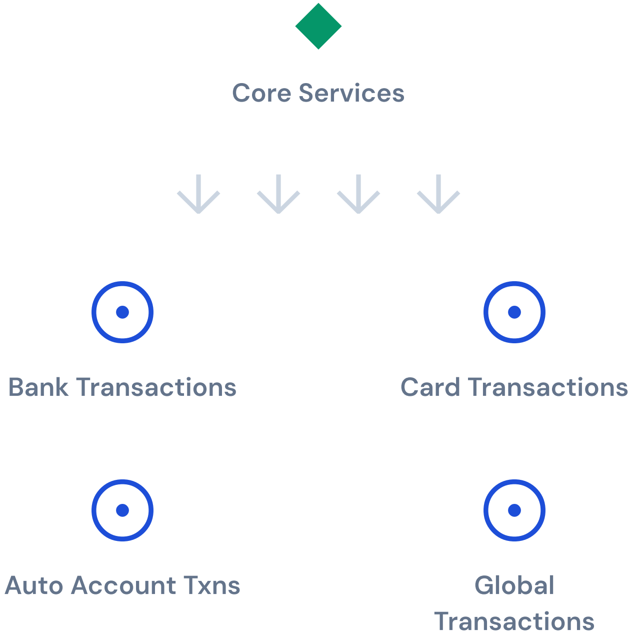


Testing Coverage

TYPE	COVERAGE	DETAIL
Sonar	98%	Full codebase
Unit	100%	Business logic, components independently
Integration	BDD	Gherkin scenarios — API, AWS, DynamoDB layers
Performance	1B+ records	Apache JMeter load testing

BDD in practice: Integration tests used Gherkin-based scenarios mirroring real-world workflows and acceptance criteria through behavior-driven development.

NoSQL + Go + Serverless. Purpose-built for scale.



NoSQL · DynamoDB · Go · Lambda · Terraform · Multi-Region

- **Business use cases** decide the entire tech stack
- Schema designed with **room for future changes**
- **DynamoDB + Go** — ideal and cost-effective at scale
- **1,000+ concurrent Lambdas** handling millions of records

> **Sign-off:** Product Owners, Third-Party Vendors, Enterprise Architects, VP Leadership. Every architectural decision required cross-functional approval.

09 · IMPACT & TAKEAWAYS

Challenge existing designs. Let the product decide.

130M+
ACCOUNTS

8 mo
DELIVERED

4
SERVICES

98%
COVERAGE

- **Product requirements define architecture.** Schema → DB → Cloud → Language. Every decision traces back.
- **Challenge inherited designs.** We moved from the default k8s path to serverless. Saved cost and complexity.
- **Root-cause analysis over gut.** Account volumes and growth projections were the foundation.
- **Ripple effect.** Other teams adopted the new patterns, including end-of-day batch processing.

> Built Terraform from scratch enabling automated cross-region failover us-east-1 to us-west-2. Designed "code-once" architecture adopted across four services on multiple AWS accounts.

10 · PERSONAL INSIGHTS

A few things beyond the code

Sports

Volleyball, Cricket, Badminton. Compete whenever I can. Team sports taught me more about coordination than any standup meeting.

Driving

Love driving through Manhattan and Jersey City. The traffic isn't relaxing, but the energy is.

Languages

English, Telugu, Tamil, Hindi, Kannada — 4+ languages. Working across global teams and time zones has always felt natural.

Exploring

New cuisines, new places. Curiosity doesn't stop when I close the laptop.

Q & A

Ok, shoot.

Happy to go deeper on any technical decisions, trade-offs, challenges, or lessons learned.

LinkedIn: saikiranreddykonda **Email:** konda.saik@northeastern.edu